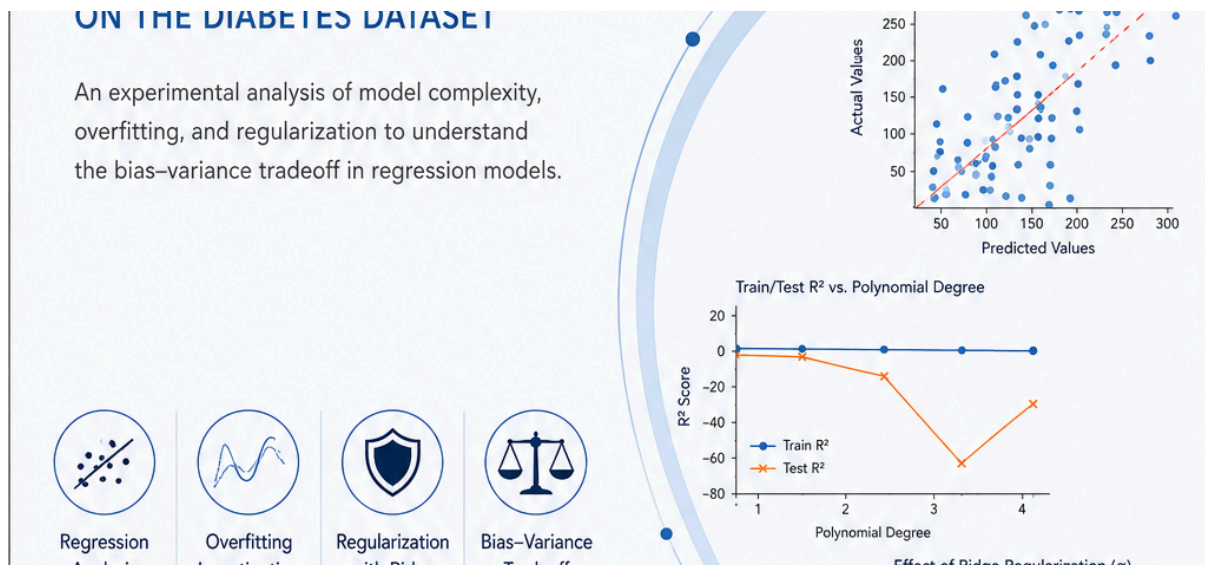




Research Hub

A Comparative Study of Linear Regression, Polynomial Regression, and Ridge Regression on the Diabetes Dataset

Introduction:

Machine learning regression models vary in their ability to capture relationships between input features and continuous target variables. While simple linear models often generalize well, increasing model complexity may improve predictive performance at the cost of overfitting. This study compares Linear Regression, Polynomial Regression, and Ridge Regression on the Diabetes dataset to investigate the trade-off between model complexity, predictive accuracy, and generalization.

Research Objectives:

The objectives of this study are:

- Establish Linear Regression as a baseline.
- Investigate the effect of increasing polynomial degree.
- Experimentally demonstrate overfitting.
- Evaluate whether Ridge Regression reduces overfitting.
- Analyze the bias–variance tradeoff.

Research Question:

How do Linear Regression, Polynomial Regression, and Ridge Regression compare on the diabetes dataset in terms of predictive performance and generalization?

Hypotheses:

H1:

Linear Regression will provide a strong baseline because the Diabetes dataset contains mostly numerical features

H2:

Polynomial Regression will improve training performance but may overfit as polynomial degree increases

H3:

Ridge Regression will reduce overfitting by shrinking coefficient magnitudes, resulting in better test performance for higher-degree polynomial models

Dataset Description:

Dataset Name:

Diabetes Dataset

Source:

```
sklearn.datasets.load_diabetes(*\, return_X_y=False, as_frame=False, scaled=True)
```

Task:

Regression

Samples:

442

Features:

10 numerical features

Target:

A quantitative measure of disease progression one year after baseline

Research Methodology:

This study follows an experimental approach to compare the performance of Linear Regression, Polynomial Regression, and Ridge Regression on the Diabetes dataset. To ensure a fair comparison, all models were trained and evaluated using the same dataset, train-test split, and evaluation metrics. The only variables changed during the experiments were the regression algorithm, polynomial degree, and Ridge regularization parameter (α).

The experimental workflow consisted of the following steps:

1. Load the Diabetes dataset from `sklearn.datasets.load_diabetes`.`
2. Perform Exploratory Data Analysis (EDA) to understand the characteristics of the dataset.
3. Split the dataset into training (80%) and testing (20%) sets using `random_state = 42`.`
4. Train a baseline Linear Regression model using a custom implementation.
5. Generate Polynomial Features of varying degrees (2, 3, and 4) and train the custom Linear Regression model.
6. Observe the effect of increasing model complexity on predictive performance.
7. Apply Ridge Regression with Degree-2 polynomial features using different regularization strengths ($\alpha = 0.01, 0.1, 1, 2, 10, \text{ and } 100$).
8. Compare all models using identical evaluation metrics.

To maintain experimental consistency, the train-test split and evaluation procedure remained unchanged throughout the study.

Evaluation Metrics:

The performance of each model was evaluated using the following regression metrics:

Coefficient of Determination (R^2)

R^2 measures the proportion of variance in the target variable explained by the model. Higher values indicate better predictive performance.

Formula:

$$R^2 = 1 - (\sum(y - \hat{y})^2 / \sum(y - \bar{y})^2)$$

where

y = Actual value

$\hat{y}$ = Predicted value

$\bar{y}$ = Mean of actual values

Mean Squared Error (MSE)

MSE measures the average squared difference between the predicted and actual values. Lower values indicate better model performance.

Formula:

$$MSE = (1/n) \sum(y - \hat{y})^2$$

Root Mean Squared Error (RMSE)

RMSE is the square root of the Mean Squared Error. It measures the average prediction error in the same units as the target variable.

Formula:

$$RMSE = \sqrt{MSE}$$

Exploratory Data Analysis:

Before training the regression models, an exploratory analysis of the Diabetes dataset was performed to understand the characteristics of the data and identify any potential issues that could affect model performance.

The analysis included:

- Dataset dimensions
- Feature information
- Descriptive statistics
- Missing value analysis
- Feature distributions
- Target distribution
- Correlation analysis
- Multicollinearity analysis

Dataset Overview:

Observation:

The dataset contains 442 observations with 10 numerical predictor variables and one continuous target variable. No missing values were found, indicating that no additional preprocessing was required. All features are numerical and suitable for regression analysis.

Feature Distribution:

Figure 1. Distribution of the Input Features and Target Variable:

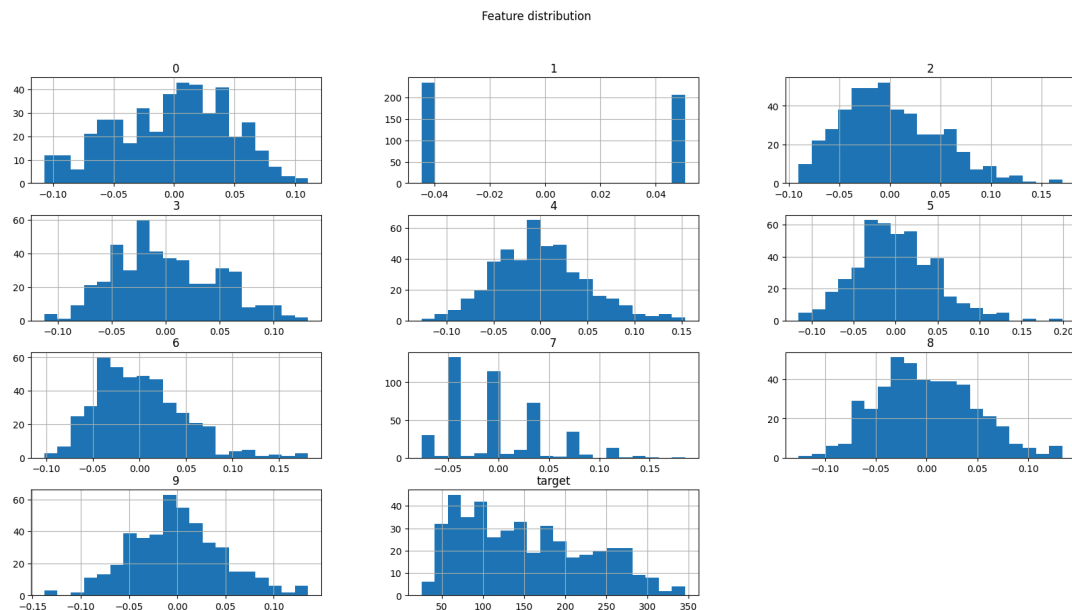


Figure 1 illustrates the distributions of all ten predictor variables together with the target variable in the Diabetes dataset

The predictor variables appear approximately centered around zero, confirming that the dataset provided by scikit-learn has already been standardized. Standardization ensures that all features contribute on a comparable scale during model training and prevents variables with larger numerical ranges from dominating the optimization process.

The target variable represents the quantitative measure of disease progression one year after baseline. Unlike the predictor variables, the target values are not standardized and exhibit a continuous distribution, making the dataset suitable for regression analysis.

No severe outliers or abnormal distributions were observed that would prevent the application of linear regression models.

Correlation Analysis:

To understand the linear relationships between predictor variables and the target variable, Pearson correlation coefficients were computed. Correlation analysis helps identify the variables that have the strongest association with the target while also revealing multicollinearity among predictor variables.

The correlation matrix and feature-target correlations are presented in Figures 2 and 3.

Figure 2. Pearson Correlation Matrix of the Diabetes Dataset:

Figure 2 presents the Pearson correlation coefficients between all predictor variables and the target variable.

BMI (feature 2) exhibits the strongest positive correlation with the target variable (approximately 0.59), followed by feature s5 (feature 8) with a correlation of approximately 0.57. Conversely, feature s3

(feature 6) demonstrates the strongest negative correlation (approximately -0.39).

The heatmap also reveals strong multicollinearity among several predictor variables. In particular, features s1 and s2 show a very high positive correlation (approximately 0.90), indicating that they contain highly similar information.

Such multicollinearity can negatively affect ordinary least squares regression by producing unstable coefficient estimates. This observation provides the primary motivation for evaluating Ridge Regression later in this study.

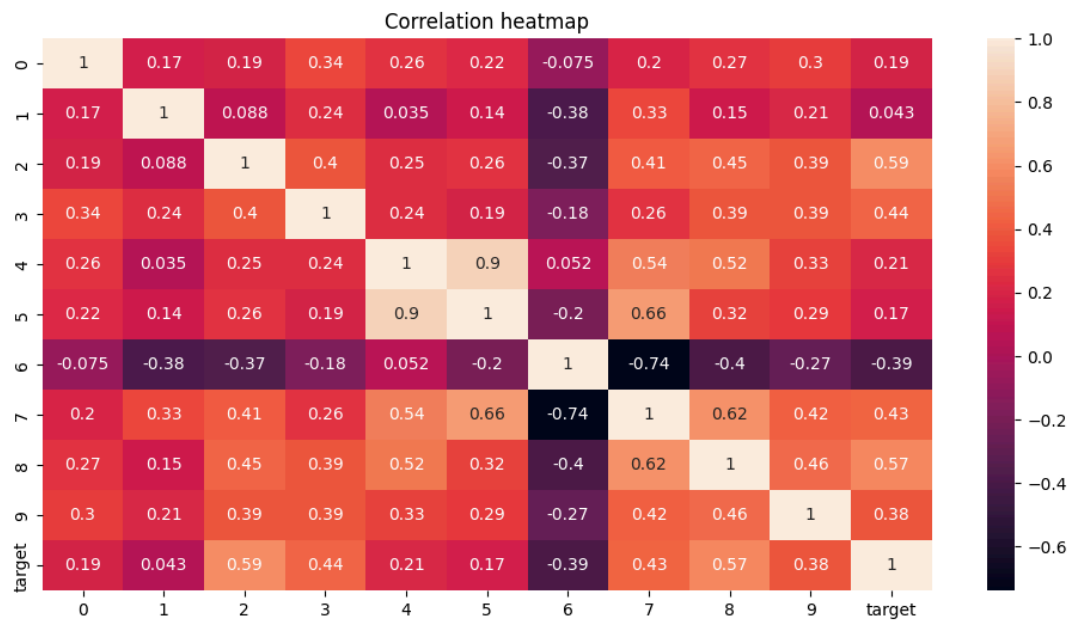


Figure 3. Correlation of Predictor Variables with the Target Variable:

Figure 1 shows the distributions of all ten predictor variables along with the target variable in the Diabetes dataset.

The predictor variables are approximately centered around zero, confirming that the scikit-learn dataset has already been standardized. Standardization ensures that all features contribute on a comparable scale during model training and prevents variables with larger numerical ranges from dominating the optimization process.

The target variable is a quantitative measure of disease progression one year after baseline. Unlike the predictors, the target values are not standardized and exhibit a continuous distribution, making the dataset suitable for regression analysis.

No severe outliers or abnormal distributions were observed that would prevent the use of linear regression models.



Transition:

Baseline Model: Linear Regression:

Following the exploratory analysis, the first experiment establishes Linear Regression as the baseline model.

The purpose of this experiment is to evaluate how well a simple linear model explains disease progression before introducing additional model complexity through polynomial feature expansion and regularization.

All subsequent models are compared against this baseline to determine whether increased complexity leads to improved predictive performance or introduces overfitting.

Experiment 1: Linear Regression

Objective:

To establish a baseline regression model using a custom implementation of Linear Regression and evaluate its predictive performance on the Diabetes dataset.

Experimental Setup:

Model:

Custom Linear Regression

Training Data:

80%

Testing Data:

20%

Random State:

42

Evaluation Metrics:

- R^2
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Results:

Metric	Train	Test
R^2	0.519	0.485
MSE	2907.26	2848.31
RMSE	53.92	53.37

Figure 4. Predicted versus Actual Values for Linear Regression

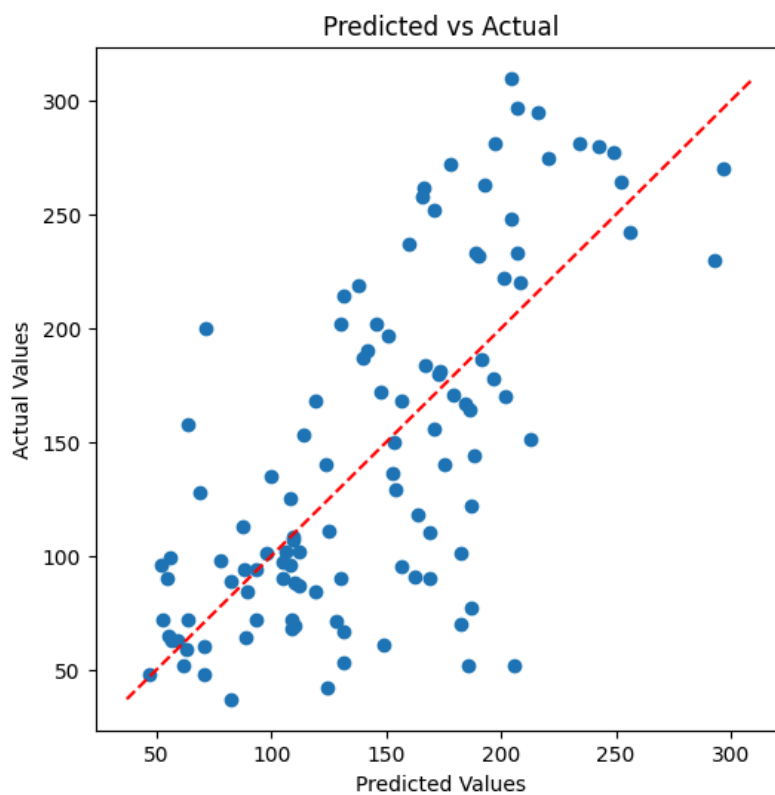


Figure 4 compares the predicted target values with the corresponding actual values for the Linear Regression model.

The red reference line represents perfect predictions, where predicted values exactly match the observed values. Most observations lie reasonably close to the reference line, indicating that the model captures the general trend of the data.

However, noticeable deviations from the reference line suggest that a simple linear relationship does not fully explain the underlying patterns within the Diabetes dataset.

Figure 5. Residual Analysis of the Linear Regression Model

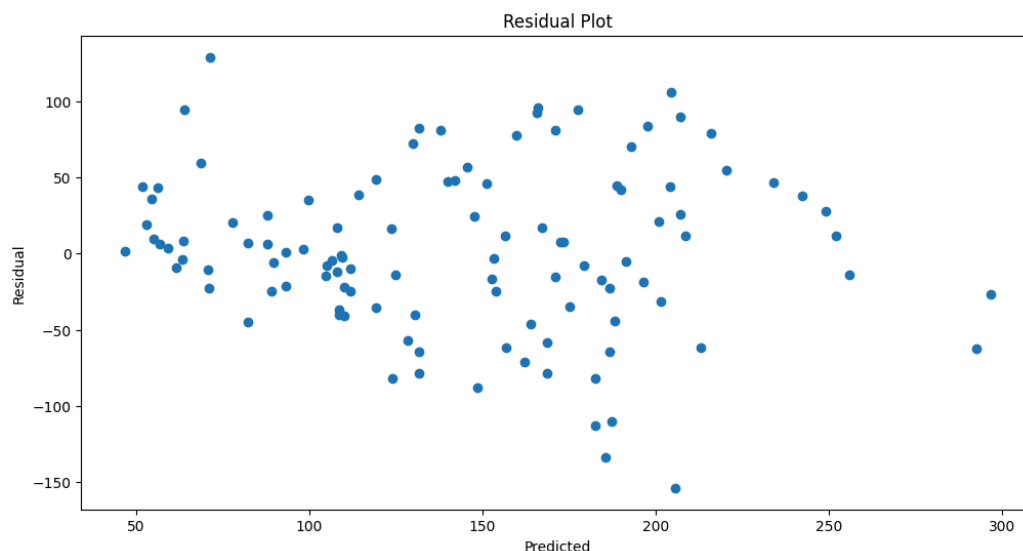


Figure 5 illustrates the residuals produced by the Linear Regression model.

The residuals are distributed around zero without a strong systematic pattern, indicating that the model captures a substantial portion of the linear relationship present in the dataset.

Nevertheless, the residual spread increases for larger predicted values, suggesting that the model is unable to capture all underlying nonlinear relationships. This observation motivates the exploration of more expressive regression models.

Interpretation:

The Linear Regression model achieved a training R^2 score of 0.519 and a testing R^2 score of 0.485. The similarity between the training and testing performance indicates that the model generalizes reasonably well and does not exhibit significant overfitting.

However, the model explains only approximately 48% of the variance in the target variable, indicating that the relationship between the predictor variables and disease progression is not entirely linear.

These findings establish Linear Regression as a strong baseline while also motivating the investigation of Polynomial Regression to determine whether increased model complexity can improve predictive performance.

Experiment 2: Polynomial Regression

Objective:

To investigate whether increasing model complexity through polynomial feature expansion improves predictive performance compared with the baseline Linear Regression model.

Polynomial Regression models of Degree 2, Degree 3, and Degree 4 were evaluated using the custom Linear Regression implementation to study the relationship between model complexity and generalization performance.

Experimental Setup

Model:

Custom Linear Regression

Polynomial Degrees:

2

3

4

Training Data:

80%

Testing Data:

20%

Random State:

42

Evaluation Metrics:

- R^2
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Results (Model Comparison Table)

Degree / Model	Features	Train R^2	Test R^2	Train RMSE	Test RMSE
2	66	0.604815329	0.424241945	48.874768258	56.42484769
3	286	0.909838	-55.93834654	23.3451016	561.11620
4	1001	1.0	-29.4147392	3.414125889060379e-11	410.1025158241

Effect of Polynomial Degree on Model Performance

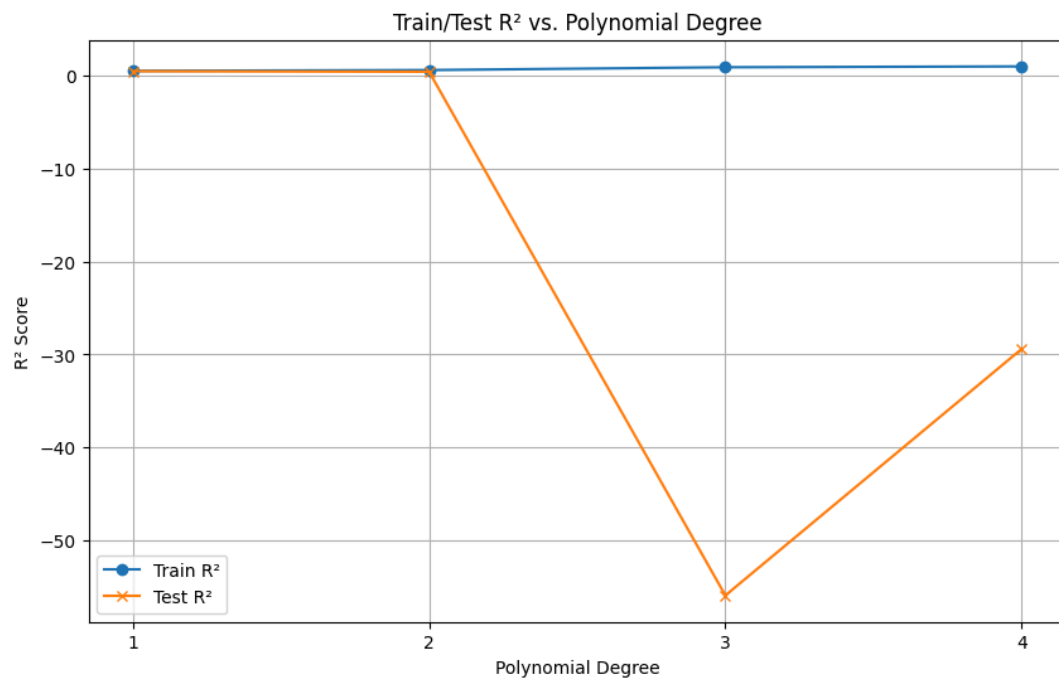


Figure 6 illustrates how polynomial degree affects the model's training and testing performance.

As the polynomial degree increases, the number of generated features grows substantially, allowing the model to capture more complex relationships in the training data. As a result, the training R^2 score rises steadily from Linear Regression (Degree 1) to higher-degree polynomial models.

However, the testing R^2 score shows a different pattern. While Degree 2 performs comparably to the baseline model, the testing R^2 drops sharply at Degree 3 and remains highly negative at Degree 4. This widening gap between training and testing R^2 indicates that the model is memorizing the training data rather than learning patterns that generalize to unseen observations.

Overall, these results highlight the effect of increasing model complexity on generalization. Although higher-degree polynomial models achieve excellent training performance, they exhibit severe overfitting and perform poorly on the test set. These findings motivate the use of Ridge Regression in the next experiment to examine whether regularization can control model complexity and improve generalization.

Interpretation:

The Polynomial Regression experiments demonstrate the relationship between model complexity and predictive performance.

Degree 2 Polynomial Regression improved the training performance compared with the baseline Linear Regression model, indicating that the additional polynomial features enabled the model to capture nonlinear relationships within the data. However, the testing performance decreased slightly, suggesting the onset of overfitting.

For Degree 3, the training R^2 score increased dramatically while the testing R^2 score became strongly negative. This indicates that the model memorized the training data but completely failed to generalize to unseen samples.

Degree 4 further amplified this behavior by achieving an almost perfect fit on the training data while producing extremely poor testing performance. The exponential increase in the number of polynomial features significantly increased model complexity, resulting in severe overfitting.

These experiments clearly demonstrate the classical bias–variance tradeoff. Increasing model complexity reduces training error but eventually causes the model to lose its ability to generalize.

The observed overfitting naturally motivates the application of Ridge Regression, which introduces L2 regularization.

Motivation for Ridge Regression

The Polynomial Regression experiments demonstrated that simply increasing model complexity does not guarantee improved predictive performance. Although higher-degree polynomial models achieved excellent training performance, they failed to generalize to unseen data because of severe overfitting.

This observation raises an important research question:

Can regularization reduce overfitting while retaining the benefits of polynomial feature expansion?

To answer this question, Ridge Regression is introduced in the next experiment.

Experiment 3: Ridge Regression

Objective

To investigate whether Ridge Regression can reduce the overfitting observed in Polynomial Regression by introducing L2 regularization.

Degree-2 Polynomial Features were selected because they exhibited moderate overfitting while maintaining reasonable predictive performance. Ridge Regression was evaluated using multiple regularization strengths to determine the optimal balance between model complexity and generalization.

Experimental Setup

Model:

Custom Ridge Regression

Polynomial Degree:

2

Regularization Parameter (α):

0.01

0.1

1

2

10

100

Training Data:

80%

Testing Data:

20%

Random State:

42

Evaluation Metrics:

- R^2
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Results (Ridge Regression Comparison Table)

α	Train R^2	Test R^2	Train RMSE	Test RMSE
0.01	0.5371473311770835	0.5050889280291726	52.83995637060834	52.313523161299436
0.1	0.5149371236445714	0.49537663053177616	54.148157743296565	52.824338121110365
1	0.42791319284620444	0.43870182398674673	58.80518666287205	55.7118012407043
2	0.35502205037099654	0.3715885498080371	62.43916695620649	58.94845532339096
10	0.15099790967423465	0.15644986167540842	71.63727316614909	68.29766170166955
100	0.020217377929379388	0.009089658750056961	76.95724867689958	74.02315046422555

Figure 7. Effect of Ridge Regularization on R^2

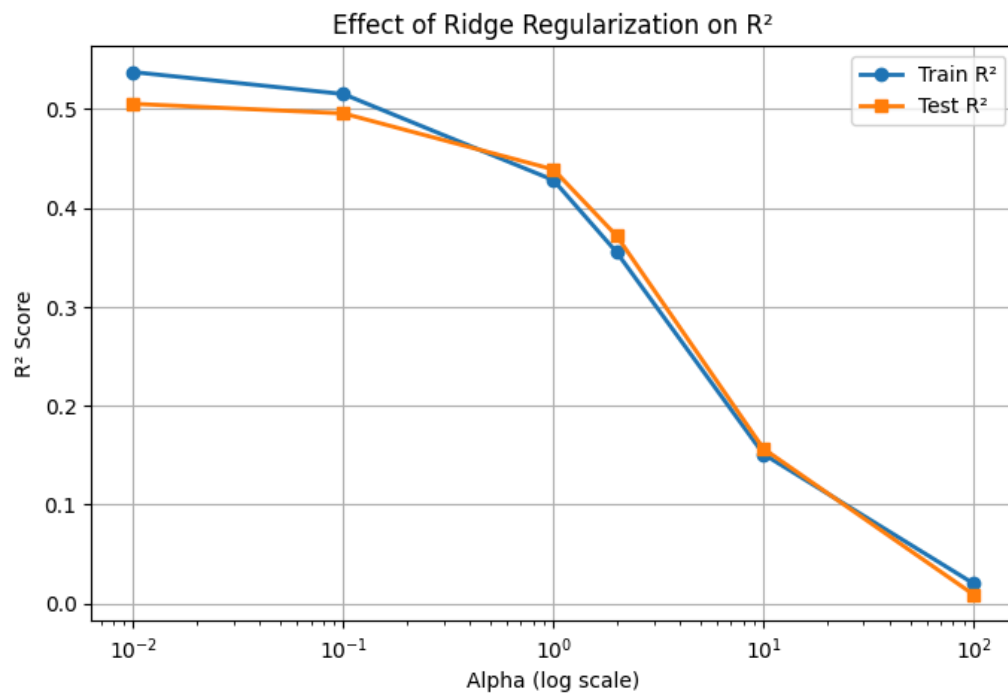


Figure 7 illustrates the effect of increasing the Ridge regularization parameter (α) on model performance.

Small values of α produced the highest testing R^2 scores while only slightly reducing training performance. This indicates that mild regularization successfully reduced overfitting without significantly increasing model bias.

As α increased, both training and testing R^2 scores steadily declined. Stronger regularization increasingly constrained the model, eventually preventing it from learning meaningful relationships from the data.

For very large α values, the model exhibited clear underfitting, as indicated by the substantial decrease in both training and testing performance.

Figure 8. Effect of Ridge Regularization on RMSE

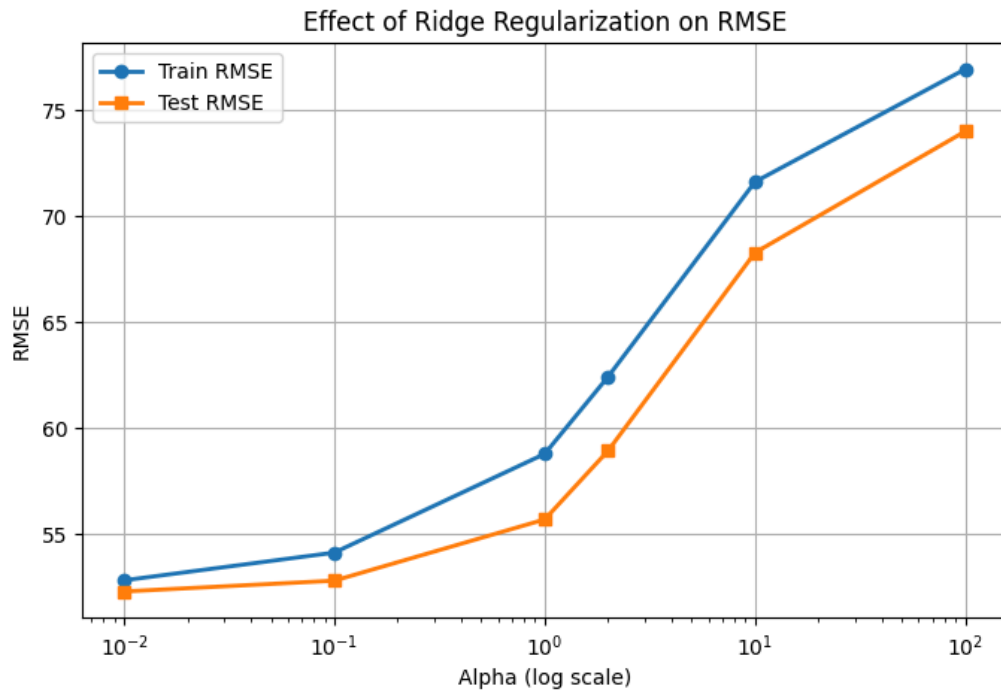


Figure 8 presents the effect of the Ridge regularization parameter on prediction error.

The lowest testing RMSE was achieved when $\alpha = 0.01$, indicating the best predictive performance among all evaluated Ridge models. Increasing α beyond this point caused both training and testing RMSE to increase steadily.

The results demonstrate that excessive regularization reduces the flexibility of the model, leading to higher prediction errors and underfitting.

Interpretation:

The Ridge Regression experiments demonstrate that the strength of regularization plays a critical role in determining predictive performance.

Very small regularization strengths ($\alpha = 0.01$ and $\alpha = 0.1$) produced the highest testing R^2 scores and the lowest testing RMSE values. These models successfully reduced overfitting while preserving the nonlinear relationships introduced by polynomial feature expansion.

As the regularization strength increased beyond $\alpha = 1$, both training and testing performance declined consistently. This behavior indicates that the increasing regularization penalty progressively restricted the model's flexibility, causing important predictive relationships to be lost.

For α values of 10 and 100, the model exhibited severe underfitting. Excessive coefficient shrinkage prevented the model from adequately fitting either the training or testing data.

Among all evaluated models, Ridge Regression with $\alpha = 0.01$ achieved the highest testing R^2 score (0.505) and the lowest testing RMSE (52.314), indicating the best balance between bias and variance for this dataset and feature expansion.

Discussion

The experiments conducted in this study illustrate the relationship between model complexity, overfitting, and regularization in supervised regression.

Linear Regression provided a stable baseline model with similar training and testing performance, indicating good generalization but limited capacity to capture nonlinear relationships.

Polynomial Regression increased model complexity by introducing higher-order feature interactions. Although this substantially improved training performance, testing performance deteriorated rapidly as the polynomial degree increased, demonstrating severe overfitting. The rapid growth in the number of generated features greatly increased the variance of the model and reduced its ability to generalize.

Ridge Regression successfully addressed this limitation by introducing L2 regularization. Small regularization strengths ($\alpha = 0.01$ and $\alpha = 0.1$) reduced model complexity sufficiently to improve testing performance while preserving the nonlinear relationships captured by the polynomial features.

However, excessively large regularization strengths produced the opposite effect. Strong coefficient shrinkage limited the model's flexibility, resulting in underfitting and reduced predictive performance.

Overall, the experiments demonstrate that neither increasing model complexity nor applying stronger regularization guarantees better performance. Instead, predictive accuracy depends on achieving an appropriate balance between model flexibility and generalization, illustrating the fundamental bias–variance tradeoff in machine learning.

Conclusion

This study compared the performance of Linear Regression, Polynomial Regression, and Ridge Regression on the Diabetes dataset to investigate the effects of model complexity and regularization. Linear Regression provided a strong baseline with good generalization, while Polynomial Regression demonstrated that increasing model complexity improved training performance but resulted in severe overfitting at higher polynomial degrees. Ridge Regression successfully mitigated overfitting through L2 regularization, with $\alpha = 0.01$ achieving the best balance between predictive accuracy and generalization. Overall, the experiments demonstrate that optimal predictive performance depends on balancing model complexity with appropriate regularization, highlighting the importance of the bias–variance tradeoff in regression modeling.